

Quiz 9

Friday, November 20 — inverses, rank, and determinants (Week 11)

Last updated: October 5, 2026.

Covers the prior instructional week only: [Inverses](#), [Rank](#), [Determinants](#) (Week 11). Midterm 2 week is skipped when choosing the prior week. Same-day [Lab 5](#) / eigenvectors are **not** on this quiz.

Textbooks: [Boyd & Vandenberghe \(VMLS\)](#) · [Strang §3.5](#) · [Lachniet Octave PDF](#) · also on [Resources](#).

[Schedule](#)

Core

- B 11.1, 11.2, 11.4, 11.10
- S 3.5 #1 and #2 (rank / subspace portions used in class)
- L Ch. 1 Ex. 3(a), 3(b), and 3(d): determinant, rank, and inverse portions
- C9.1: Decide whether a small matrix is invertible and explain via determinant or solving $Ax = b$.
- C9.2: Find rank and a basis for the column space or null space from a small RREF.
- C9.3: Compute a 2×2 or triangular determinant and interpret area/volume scaling.

Stretch

B 11.15, 11.25; C-M2-style area scaling from $\det A$.

What a quiz item might change

A new 2×2 matrix, an RREF with a free variable, or whether $\det A = 0$ in context.